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OPTICAL CONCENTRATION MEASURING DEVICE AND SIGNAL PROCESSOR

Abstract

An optical concentration measuring device includes a light source, detectors (**21, 22**) that detect a signal, based on light emitted from the light source, and output a detection signal, a signal processor (**50**) that acquires the detection signal and converts the detection signal to a digital signal, and a substrate (**30**) on which the detector and the signal processor are provided. The signal processor has a terminal array in which a plurality of signal terminals is sandwiched between a plurality of voltage terminals, the signal terminals being input and output terminals of signal lines electrically connected to the detectors, **10** the voltage terminals supplying a voltage other than 0 V. In plan view of the substrate from the front, the signal processor and wiring connecting the plurality of voltage terminals surround the signal lines.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Japanese Patent Application No. 2024-034107 filed on Mar. 6, 2024, and Japanese Patent Application No. 2025-004275 filed on Jan. 10, 2025, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an optical concentration measuring device and a signal processor.

BACKGROUND

[0003] In recent years, gas sensors (gas measurement devices) that include a light source emitting infrared light and are configured so that the infrared light passes through a gas containing a gas to be detected have been developed. These gas sensors detect the concentration of the gas to be detected by using the absorption characteristics of the infrared light by the gas to be detected. For example, infrared sensor modules that include an infrared sensor, an integrated circuit (IC) that processes the output signal of the infrared sensor, and a package housing the infrared sensor and IC have been proposed. In small infrared sensor modules, the IC may perform signal processing based on minute currents of the order of nA, for example, making it necessary to suppress leakage currents and the like to improve detection accuracy. For example, Patent Literature (PTL) 1 discloses an output current detection IC, for a diode-type sensor, in which the effect of leakage current is reduced.

CITATION LIST

Patent Literature

[0004] PTL 1: WO2015/105048A1

SUMMARY

[0005] Here, the technology in PTL 1 reduces the leakage current of a protection circuit, and it is difficult to apply this technology directly to normal measurement by gas sensors. In addition, a conventional method is known for suppressing leakage current by using a layout that surrounds the infrared sensor with a ground potential on the gas sensor substrate. The conventional method has limited effectiveness in suppressing leakage current, however, and demand exists for a more effective method.

[0006] It would be helpful to provide an optical concentration measuring device and a signal processor with an enhanced leakage current suppression effect.

[0007] (1) An optical concentration measuring device according to an embodiment of the present disclosure includes: [0008] a light source configured to emit light toward a gas detection space into which a gas to be detected is introduced; [0009] a detector configured to detect a signal, based on at least a portion of the light that was emitted from the light source and passed through the gas to be detected, and output a detection signal; [0010] a signal processor configured to acquire the detection signal and convert the detection signal to a digital signal; and [0011] a substrate on which the detector and the signal processor are provided, wherein [0012] the detector includes a first detector and a second detector, [0013] the signal processor is configured to form a terminal array in which a plurality of signal terminals is sandwiched between a plurality of voltage terminals, the

signal terminals being input and output terminals of signal lines electrically connected to the detector, the voltage terminals supplying a constant voltage other than 0 V and not being electrically connected to the detector, [0014] the plurality of voltage terminals includes a first voltage terminal, a second voltage terminal, and a third voltage terminal, and [0015] in plan view of the substrate from a front, the signal processor and a wiring connecting the first voltage terminal and the second voltage terminal surround signal lines connecting the first detector and the signal processor, and the signal processor and a wiring connecting the second voltage terminal and the third voltage terminal surround signal lines connecting the second detector and the signal processor. [0016] (2) As an embodiment of the present disclosure, in (1), the plurality of signal terminals includes at least two signal terminals, for connection to the first detector, sandwiched between the first voltage terminal and the second voltage terminal, and at least two signal terminals, for connection to the second detector, sandwiched between the second voltage terminal and the third voltage terminal. [0017] (3) As an embodiment of the present disclosure, in (2), the plurality of signal terminals includes two signal terminals, for connection to the first detector, sandwiched between the first voltage terminal and the second voltage terminal, and two signal terminals, for connection to the second detector, sandwiched between the second voltage terminal and the third voltage terminal. [0018] (4) As an embodiment of the present disclosure, in any one of (1) to (3), in plan view of the substrate from the front, the signal processor is rectangular in shape, and the plurality of voltage terminals and the plurality of signal terminals are all provided on one side of the signal processor. [0019] (5) As an embodiment of the present disclosure, in any one of (1) to (4), a portion of the wiring connecting the first voltage terminal and the second voltage terminal is common to a portion of the wiring connecting the second voltage terminal and the third voltage terminal. [0020] (6) As an embodiment of the present disclosure, in any one of (1) to (5), a wiring length from the signal processor to the first detector is shorter than a wiring length from the signal processor to the second detector. [0021] (7) As an embodiment of the present disclosure, in any one of (1) to (6), a portion of the wiring has a disconnected section, and a length of the disconnected section is 1/10 or less of a length of a signal line connecting to the detector from the signal terminal. [0022] (8) A signal processor according to an embodiment of the present disclosure is useable in the optical concentration measuring device of any one of (1) to (7). [0023] According to the present disclosure, an optical concentration measuring device and a signal processor with an enhanced leakage current suppression effect can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] In the accompanying drawings:

[0025] FIG. 1 is a diagram illustrating an example configuration of a gas sensor according to an embodiment of the present disclosure; and

[0026] FIG. 2 is a plan view of a substrate from the front, with a portion of the gas sensor in FIG. 1 being transparent.

DETAILED DESCRIPTION

[0027] The gas sensor **10** (see FIG. 1) and the signal processor **50** (see FIG. 2) according to an embodiment of the present disclosure are described below with reference to the drawings.

(Gas Sensor **10**)

[0028] FIG. 1 is a diagram illustrating a configuration of the gas sensor **10** according to the present embodiment. The gas sensor **10** measures the presence or concentration of a gas to be detected included in a gas. In the present embodiment, the gas sensor **10** is described as a device that

measures the concentration of the gas to be detected. The gas sensor **10** in the present embodiment is an NDIR (non-dispersive infrared) gas sensor, but the gas sensor **10** is not limited to this configuration and may, for example, be a photoacoustic gas sensor. The NDIR gas sensor **10** measures the concentration of the gas to be detected using an infrared light receiving element that receives infrared light in an absorption wavelength band corresponding to the gas to be detected and an infrared light emitting element that emits infrared light in that absorption wavelength band. In the case of the gas sensor **10** being a photoacoustic gas sensor, the concentration of the gas to be detected is measured by using a high-performance microphone to pick up, as sound (pressure change), the vibration of the gas molecules that absorbed the light. The gas sensor **10** in the present embodiment can be applied to a variety of devices. For example, the gas sensor **10** can be used for environmental measurement in buildings, mounted on portable communication devices such as smartphones as a portable compact measuring device, or used for gas detection in the interior of vehicles such as cars, trains, or airplanes.

[0029] According to the configuration of the gas sensor **10** in the present embodiment, the gas sensor **10** is applicable as a light receiving/emitting device for applications other than gas detection. That is, disclosure content derived by replacing “gas sensor **10**” as described above with “optical concentration measuring device”, “optical physical quantity measuring device”, “light receiving/emitting device”, “optical device”, or the like is included in the scope of the present disclosure. For example, the state of an optical path space can be detected (examples other than gas include the presence or absence or concentration of a specific component of a fluid). For example, the disclosure content can be used for a component detection device or a component concentration measuring device for a substance (for example, water or a body fluid) present in an optical path space between a light emitter and a light receiver. For example, when the substance present in the optical path space is blood, the component detection device or the component concentration measuring device can be used to measure glucose concentration in blood.

[0030] The component detection device or the component concentration measuring device can measure glucose concentration in blood sugar by measuring absorption of light having a wavelength of 1 μm to 10 μm . In the measurement of glucose concentration in blood sugar, measuring absorption of light in the 1.6 μm band, the 2.0 μm band, and the 10.0 μm band is preferred. A compact, high precision, and highly reliable non-invasive glucose concentration meter can be realized. Such a glucose concentration meter allows, for example, a diabetic patient to self-check blood sugar levels with good precision and without causing damage to the skin as would occur with an invasive method. Furthermore, more accurate administration of medication (for example, insulin) can be achieved, based on the blood sugar levels checked.

[0031] The gas sensor **10** includes a light source **11**, a detector **20**, a signal processor **50** (see FIG. 2), and a substrate **30**. In the present embodiment, the detector **20** is configured by a first detector **21** and a second detector **22**. The gas sensor **10** also includes a gas cell **40**. Here, the detector **20** detects a signal based on light emitted from the light source **11** and outputs a detection signal. The signal detected by the detector **20** may be the light itself (the aforementioned NDIR method) or may include vibrations of gas molecules that have absorbed the light (the aforementioned photoacoustic method).

[0032] The gas cell **40** has a hole **43** through which a gas (for example, ambient air) passes, and gas is introduced through the hole **43** into a gas detection space **42**. The gas sensor **10** measures the presence or concentration of a gas to be detected included in the introduced gas. The gas to be detected can be a combustible gas such as carbon dioxide, exhaled alcohol (ethanol or the like), methane, propane, hydrogen, ethylene, or methylcyclohexane (MCH). The gas to be detected may also be a toxic gas such as carbon monoxide, hydrogen sulfide, formaldehyde, or ammonia. Furthermore, the gas to be detected may be a greenhouse gas, such as dinitrogen monoxide or refrigerant gases used in air conditioners or refrigerators (freon, freon substitutes, R32, R1234y). The gas sensor **10** measures the concentration of the gas to be detected in the gas introduced into

the gas detection space **42** and outputs the measurement results.

[0033] The gas sensor **10** is configured to include the gas cell **40** on a main surface **30a** of the substrate **30**. Here, the main surface **30a** is one of the faces with the largest area among the faces of the substrate **30**. As illustrated in FIG. **1**, Cartesian coordinates are set so that the xy-plane is parallel to the main surface **30a** of the substrate **30**. The z-axis direction is a direction perpendicular to the main surface **30a** of the substrate **30**. The z-axis direction is also referred to as the up-down direction. In this case, the positive z-axis direction corresponds to the up direction. The plan view is the view looking in the negative z-axis direction (down direction). In the present embodiment, the light source **11**, the detector **20**, and the signal processor **50** are provided on the main surface **30a** of the substrate **30** and are arranged inside the gas detection space **42** on the main surface **30a**.

[0034] In the present embodiment, the second detector **22** is integrated with the light source **11**. In greater detail, the light source **11** and the second detector **22** are sealed on one side of a light source substrate **12** by a seal **23**. In other words, the light source substrate **12**, the light source **11**, the second detector **22**, and the seal **23** are integrated to form a light source module. FIG. **1** illustrates a cross-section view of the gas sensor **10** including these components. Here, a detection substrate may be present for the first detector **21** on the light input side in correspondence with the light source substrate **12**, but such a detection substrate is omitted from FIG. **1** for clarity.

[0035] FIG. **2** is a plan view of the substrate **30** from the front, with a portion of the gas sensor **10** in FIG. **1** being transparent. In greater detail, FIG. **2** is a plan view of the substrate **30** having the detector **20** (first detector **21** and second detector **22**) and the signal processor **50** provided thereon, with the gas cell **40**, the light source substrate **12**, the light source **11**, and the seal **23** being transparent. The components of the gas sensor **10** are described in detail below, followed by a description of the layout of the signal processor **50** and substrate **30** illustrated in FIG. **2**.

(Substrate **30**)

[0036] The substrate **30** functions to hold the gas cell **40**, the light source module, the first detector **21**, and the signal processor **50**. In the present embodiment, the substrate **30** is a printed circuit board (PCB) for mounting the light source module, the first detector **21**, and the signal processor **50** and electrically connecting these components. The material of the substrate **30** is, for example, paper, glass cloth, ceramic, polyimide, or liquid crystal polymer.

(Gas Cell **40**)

[0037] The gas cell **40** has the gas detection space **42** for detecting the gas to be detected, as described above. The gas detection space **42** is an interior space enclosed by the outer wall of the gas cell **40** and the substrate **30**. The material of the outer wall of the gas cell **40** can, for example, be metal, glass, resin, or a composite of these materials. The resin can, for example, be phenolic resin, epoxy resin, polyimide resin, bismaleimide triazine resin, fluorine resin, polyphenylene oxide resin, liquid crystal polymer (LCP), polypropylene (PP), or polyetheretherketone (PEEK). The resin may also be polyamide (PA), polyphenylene ether (PPE), polycarbonate (PC), polyphenylene sulfide (PPS), polymethyl methacrylate resin (PMMA), or polyarylate resin (PAR). The resin may also be a hard resin or the like in which two or more of these materials are mixed. Furthermore, the gas cell **40** is preferably configured by a material with a low light absorption coefficient and high reflectance in order to efficiently confine the light emitted from the light source **11** in the gas cell **40**. Specifically, the material of the gas cell **40** is preferably a resin housing coated by an alloy containing aluminum, gold, or silver, a dielectric, or a laminate of these materials. From the perspective of high productivity and light weight, the inner surface of the gas cell **40** is preferably formed by vapor deposition or plating on the resin housing. The gas cell **40** may be formed by cutting, but from the perspective of productivity, the gas cell **40** is preferably formed by injection molding or pressing. The gas cell **40** and substrate **30** may be mechanically joined by adhesive, screws, tabs, fittings, grommets, welding, solder, or the like.

[0038] Here, in a case in which the gas cell **40** is made of a conductive material such as metal, and the substrate **30** and the gas cell **40** are electrically connected by solder or the like, the gas cell **40**

may be fixed to a reference potential. By being fixed to the reference potential, the gas cell **40** can function as an electromagnetic shield and enhance measurement performance for the gas to be detected. The reference potential is, for example, a ground potential.

(Light Source **11**)

[0039] The light source **11** emits light that contains wavelengths absorbed by the gas to be detected. The light source **11** is also referred to as a light emitter. The light source **11** may be provided on the main surface **30a** of the substrate **30** inside the gas detection space **42**. In the present embodiment, the light is infrared light. That is, the light source **11** is configured by a light emitting element that emits infrared light. The light emitting element is preferably driven at high speed by voltage pulses or current pulses. The light emitting element may, for example, be a Light Emitting Diode (LED), a Light Amplification by Stimulated Emission of Radiation (laser), a MEMS heater, or a lamp. LEDs include, for example, resonant light-emitting diodes. Lasers include, for example, vertical-cavity surface-emitting lasers. In the present embodiment, the light source **11** is an LED that emits infrared light (infrared LED). The light source **11** is preferably a quantum infrared LED that emits infrared light using electrons or holes in a semiconductor so that the light source **11** can be driven at a high frequency.

[0040] Here, the gas sensor **10** may further include a drive source that outputs a drive signal to drive the light emitting element. In the present embodiment, the drive source is included in the signal processor **50**. That is, a portion of the signal processor **50** functions as the drive source.

[0041] The light source **11** may include auxiliary members that have auxiliary optical functions such as wavelength selection, light focusing, scattering, and wavelength conversion. Specifically, the auxiliary members are wavelength selective filters, lenses, phosphors, diffraction gratings, and the like.

[0042] The light source **11** emits light from a light emitting surface **11a**. The light emitting surface **11a** is the surface of the light source **11** that is in contact with the light source substrate **12**. The light emitting surface **11a** corresponds to one of the faces of the light source **11** that has the largest area.

[0043] The wavelength of infrared light may be between 2 μm and 12 μm . The region of 2 μm to 12 μm is a particularly suitable wavelength range for use in the gas sensor **10**, as many absorption bands specific to various gases exist in this region. For example, absorption bands exist for methane at a wavelength of 3.3 μm , carbon dioxide at a wavelength of 4.3 μm , and alcohol (ethanol) at a wavelength of 9.5 μm .

[0044] A portion of the light emitted by the light source **11** may be directed toward the gas detection space into which the gas to be detected is introduced. Other portions of the light emitted by the light source **11** need not be directed toward the gas detection space into which the gas to be detected is introduced. The light emitted by the light source **11** may directly be directed toward the gas detection space into which the gas to be detected is introduced, or the light may be directed toward the gas detection space into which the gas to be detected is introduced via a reflector (light guide) such as a mirror.

(Seal **23**)

[0045] The seal **23** is formed by a mold resin, for example, but this configuration is not limiting, as long as the seal **23** seals the light source **11** and the second detector **22**. In a case in which the seal **23** is formed by mold resin, the light source **11** and the second detector **22** can be integrated with the light source substrate **12** while maintaining the positional relationship between the light source **11** and the second detector **22**. Epoxy resin, phenolic resin, or the like may, for example, be used as the material of the mold resin.

(Light Source Substrate **12**)

[0046] The light source substrate **12** may, for example, be a semiconductor substrate. The semiconductor substrate may, for example, be a Si substrate, an InP substrate, or a GaAs substrate. The light source substrate **12** may include auxiliary members that have auxiliary optical functions

such as wavelength selection, light focusing, scattering, and wavelength conversion. Specific examples of the auxiliary members are wavelength selective filters, lenses, phosphors, diffraction gratings, and the like.

(Second Detector 22)

[0047] The second detector **22** functions as a photodetector to detect the amount of light emitted from the light source **11**. The second detector **22** may be configured to include a light receiving element that is sensitive to the wavelength band of the light emitted from the light source **11**. As a specific example, the light receiving element may be a quantum type sensor, such as a photodiode with a PIN structure. As another example, the light receiving element may be a phototransistor, a thermopile, a pyroelectric sensor, a bolometer, or the like. In the present embodiment, the light receiving element is a quantum infrared sensor. The second detector **22** is located at the position of incidence of light, among the light emitted from the light source **11**, that is reflected at the surface of the light source substrate **12** in contact with the gas detection space **42** and that passes through the interior of the light source substrate **12**. By being arranged at such a position, the second detector **22** can detect the amount of light emitted from the light source **11** regardless of the usage environment.

[0048] Here, the gas sensor **10** has an analog-to-digital converter that converts the analog signal (detection signal) outputted by the second detector **22** to a digital signal. The analog signal (detection signal) outputted by the first detector **21** is also converted to a digital signal. In the present embodiment, the analog-to-digital converter is included in the signal processor **50**. In other words, the signal processor **50** has the function of acquiring the detection signal outputted from the detector **20** and converting the detection signal to a digital signal.

(First Detector 21)

[0049] During concentration measurement of the gas to be detected that is present in the gas (air) in the gas detection space **42**, the first detector **21** detects changes in accordance with the amount of the gas to be detected that is present. The first detector **21** is provided on the main surface **30a** of the substrate **30** inside the gas detection space **42**. In the present embodiment, the first detector **21** may be configured to include a light receiving element that is sensitive to the wavelength band of the light emitted from the light source **11**. As a specific example, the light receiving element may be a quantum type sensor, such as a photodiode with a PIN structure. As another example, the light receiving element may be a phototransistor, a thermopile, a pyroelectric sensor, a bolometer, or the like. In the present embodiment, the light receiving element is a quantum infrared sensor. The first detector **21** is located at the position of incidence of light, among the light emitted from the light source **11**, that passes through the gas detection space **42**. Here, the gas detection space **42** may be provided with a reflective portion (light guiding portion) that reflects light so that the light emitted from the light source **11** is reflected to be incident on the first detector **21**. The reflective portion may, for example, be a concave mirror, the reflective surface of which may be formed by a metal having a high reflectance, such as aluminum or gold.

[0050] The first detector **21** may receive light, among the light emitted from the light source **11**, that has passed through the gas detection space into which the gas to be detected is introduced. Among the light emitted from the light source **11**, the first detector **21** may receive both light that has passed through the gas detection space into which the gas to be detected is introduced and light that has not passed through the gas detection space into which the gas to be detected is introduced.

[0051] As another example, in a case in which the gas sensor **10** is photoacoustic, the first detector **21** can, for example, be a microphone, a barometric pressure sensor, or a pressure sensor.

(Gas Detection Space 42)

[0052] The gas detection space **42** has a space separated by outer walls and functions to contain gases such as air in this interior space. The gas contained in the gas detection space **42** is replaced through the hole **43**.

(Hole 43)

[0053] The hole **43** is provided in a portion of the outer walls (side walls and ceiling) that define the gas detection space **42** of the gas cell **40**. Gas passes through the hole **43** and replaces the gas inside the gas detection space **42**. More than one hole **43** may be provided. Here, the hole **43** may be provided with a dust filter to prevent dust. The dust filter may, for example, be a non-woven fabric, a Teflon® sheet (Teflon is a registered trademark in Japan, other countries, or both), or the like.

(Signal Processor **50**)

[0054] The signal processor **50** may control the entire gas sensor **10**. For example, the signal processor **50** may perform calculations to obtain the concentration of the gas to be detected. In other words, the signal processor **50** may calculate the concentration of the gas to be detected based on the detection signals from the detector **20**. As described above, in the present embodiment, the signal processor **50** has at least the function of acquiring the detection signal outputted from the detector **20** and converting the detection signal to a digital signal. In the present embodiment, the signal processor **50** also functions as a drive source to drive the light emitting element. The signal processor **50** may be configured to include one or more processors. The processor may, for example, be a general-purpose processor or a dedicated processor specialized for particular processing, but the processor is not limited to these examples and may be any processor. The signal processor **50** may be a compact signal processing device configured by an IC that includes one or more processors.

(Design of Gas Sensor **10**)

[0055] Here, to effectively suppress leakage current and improve the measurement accuracy of the gas to be detected, the gas sensor **10** according to the present embodiment adopts the layout of the signal processor **50** and substrate **30** as illustrated in FIG. 2.

[0056] First, the signal processor **50** has a terminal array in which a plurality of signal terminals, which are input and output terminals of signal lines (thin solid lines in FIG. 2) connected to the detector **20**, is sandwiched between a plurality of voltage terminals that supply voltage. The voltage supplied by the plurality of voltage terminals may be a constant voltage other than 0 V. The constant voltage here includes cases in which the voltage can be considered approximately constant when taking into consideration errors such as voltage fluctuations caused by the operation of peripheral circuits, voltage fluctuations caused by device variations, voltage fluctuations caused by temperature characteristics, and the like. In the example in FIG. 2, the plurality of voltage terminals are configured by a first voltage terminal (TP1), a second voltage terminal (TP2), and a third voltage terminal (TP3). In the present embodiment, the plurality of voltage terminals is not at ground potential (not 0 V) and provides an analog reference voltage used within the analog signal processing circuit. The analog signal processing circuit amplifies the detection signal, which is an analog signal, or reduces noise. In the example in FIG. 2, the plurality of signal terminals are configured by two signal terminals (TS1, TS2) for connection with the first detector **21** and two signal terminals (TS3, TS4) for connection with the second detector **22**. The two signal terminals (TS1, TS2) are sandwiched between the first voltage terminal (TP1) and the second voltage terminal (TP2). The two signal terminals (TS3, TS4) are sandwiched between the second voltage terminal (TP2) and the third voltage terminal (TP3). Here, it suffices for the number of signal terminals sandwiched between the first voltage terminal (TP1) and the second voltage terminal (TP2) to be at least two, and this number may, for example, be three or more. Similarly, it suffices for the number of signal terminals sandwiched between the second voltage terminal (TP2) and the third voltage terminal (TP3) to be at least two, and this number may, for example, be three or more.

[0057] In the gas sensor **10**, in plan view, the signal processor **50** and wiring connecting the plurality of voltage terminals (thick solid line in FIG. 2) surround the signal lines (thin solid lines in FIG. 2) connecting the detector **20** and the signal processor **50**. In other words, the wiring layout of the substrate **30** is designed so that the signal lines (thin solid lines in FIG. 2) are shielded by the wiring (thick solid lines in FIG. 2) of the analog reference voltage. In the example in FIG. 2, in

plan view, the signal processor **50** and the wiring connecting the first voltage terminal (TP1) and the second voltage terminal (TP2) surround the signal lines connecting the first detector **21** and the signal processor **50**. The signal processor **50** and the wiring connecting the second voltage terminal (TP2) and the third voltage terminal (TP3) surround the signal lines connecting the second detector **22** and the signal processor **50**. Here, “surround” does not mean that the wiring connecting the plurality of voltage terminals (thick solid lines in FIG. 2) must completely enclose the signal lines (thin solid lines in FIG. 2). It suffices for “surround” to mean that the signal lines (thin solid lines in FIG. 2) are enclosed by the signal processor **50** and the wiring connecting the plurality of voltage terminals (thick solid lines in FIG. 2). A portion of the wiring that surrounds the signal lines and connects the plurality of voltage terminals may have a disconnected section. The disconnected section is a portion where the wiring is not connected (discontinuous portion). However, to effectively suppress leakage current and improve the accuracy of measurement of the gas to be detected, the length of the disconnected section is preferably 1/10 or less of the length of the signal line connecting to the detector **20** from the signal terminal. In FIG. 2, the wiring connecting the plurality of voltage terminals (thick solid lines in FIG. 2) is indicated by a dashed line at the portion where the wiring overlaps the first detector **21** or the second detector **22**. The wiring in the portion indicated by the dashed line signifies that the wiring passes below (in the negative z-axis direction) the first detector **21** or the second detector **22** in plan view of the substrate **30** from the front. As another example, the wiring connecting the plurality of voltage terminals may extend in the xy-plane direction so as to pass around the first detector **21** or the second detector **22**, rather than passing below (in the negative z-axis direction) the first detector **21** or the second detector **22**. [0058] Letting the voltage of the signal line connecting the detector **20** and the signal processor **50** be V_{sig} and the analog reference voltage be V_{ref} , leakage current is likely to flow when the difference between V_{sig} and V_{ref} is large. R in FIG. 2 represents the insulation resistance of the substrate **30**. The leakage current is given by $|V_{ref} - V_{sig}|/R$. R is generally on the order of gigaohms, but the interconnections of the substrate **30** are not completely insulated. Here, during operation of the gas sensor **10** (i.e., during measurement of the gas to be detected), the value of V_{sig} (i.e., the level of the detection signal outputted by the detector **20**) is closer to V_{ref} than to 0 V. Therefore, in the configuration of the gas sensor **10** according to the present embodiment, the leakage current during operation of the gas sensor **10** can be suppressed because the value of $|V_{ref} - V_{sig}|$ is close to zero during operation of the gas sensor **10**.

[0059] For example, in conventional technology, the detector **20** is surrounded by a ground potential. In such conventional technology, a leakage current given by V_{sig}/R is generated during operation of the gas sensor **10**. Therefore, the gas sensor **10** according to the present embodiment can effectively suppress the leakage current as compared to conventional technology.

[0060] Here, in plan view of the example in FIG. 2, the signal processor **50** is rectangular in shape, and the plurality of voltage terminals and the plurality of signal terminals are all provided on one side of the signal processor **50**. The signal processor **50** may, however, be configured so that a portion of the plurality of voltage terminals and the plurality of signal terminals is provided on another side. Depending on the arrangement of the plurality of voltage terminals and the plurality of signal terminals, it may be difficult to make the distance from the signal processor **50** to the first detector **21** and the distance from the signal processor **50** to the second detector **22** the same. In this case, it is preferable to arrange the first detector **21**, which is directly related to the measurement of the concentration of the gas to be detected, to be closer to the signal processor **50** than the second detector **22**, as illustrated in FIG. 2, for example. In other words, the wiring length from the signal processor **50** to the first detector **21** is preferably shorter than the wiring length from the signal processor **50** to the second detector **22**. This is to ensure that the effect of noise on the detection signal from the first detector **21** is as small as possible.

[0061] In order to reduce the size of the gas sensor **10**, a portion of the wiring connecting the first voltage terminal (TP1) and the second voltage terminal (TP2) is preferably common to a portion of

the wiring connecting the second voltage terminal (TP2) and the third voltage terminal (TP3).
[0062] As described above, the optical concentration measurement device and the signal processor 50 according to the present embodiment can, with the aforementioned configuration, enhance the effect of suppression of leakage current during measurement.

[0063] Although embodiments of the present disclosure have been described through drawings and examples, it is to be noted that various changes and modifications will be apparent to those skilled in the art based on the present disclosure. Therefore, such changes and modifications are to be understood as included within the scope of the present disclosure.

[0064] A configuration with two detectors 20 has been described in the above embodiment, but there may be only one detector 20 (the first detector 21), or there may be three or more. In addition, in the case of the detector 20 having a plurality of configurations in the above embodiment, there is a common voltage terminal (i.e., TP2) across the signal terminals for connection to each detector, but a configuration with no common voltage terminal may be adopted. In other words, in the above embodiment, the signal processor 50 has a terminal array with seven terminals (TP1, TS1, TS2, TP2, TS3, TS4, TP3). Here, TP2 may be separated into two terminals (TP2A and TP2B), so that the signal processor 50 has a terminal array with eight terminals (TP1, TS1, TS2, TP2A, TP2B, TS3, TS4, TP3).

Claims

1. An optical concentration measuring device comprising: a light source configured to emit light toward a gas detection space into which a gas to be detected is introduced; a detector configured to detect a signal, based on at least a portion of the light that was emitted from the light source and passed through the gas to be detected, and output a detection signal; a signal processor configured to acquire the detection signal and convert the detection signal to a digital signal; and a substrate on which the detector and the signal processor are provided, wherein the detector comprises a first detector and a second detector, the signal processor is configured to form a terminal array in which a plurality of signal terminals is sandwiched between a plurality of voltage terminals, the signal terminals being input and output terminals of signal lines electrically connected to the detector, the voltage terminals supplying a constant voltage other than 0 V and not being electrically connected to the detector, the plurality of voltage terminals comprises a first voltage terminal, a second voltage terminal, and a third voltage terminal, and in plan view of the substrate from a front, the signal processor and a wiring connecting the first voltage terminal and the second voltage terminal surround signal lines connecting the first detector and the signal processor, and the signal processor and a wiring connecting the second voltage terminal and the third voltage terminal surround signal lines connecting the second detector and the signal processor.
2. The optical concentration measuring device according to claim 1, wherein the plurality of signal terminals comprises at least two signal terminals, for connection to the first detector, sandwiched between the first voltage terminal and the second voltage terminal, and at least two signal terminals, for connection to the second detector, sandwiched between the second voltage terminal and the third voltage terminal.
3. The optical concentration measuring device according to claim 2, wherein the plurality of signal terminals comprises two signal terminals, for connection to the first detector, sandwiched between the first voltage terminal and the second voltage terminal, and two signal terminals, for connection to the second detector, sandwiched between the second voltage terminal and the third voltage terminal.
4. The optical concentration measuring device according to claim 1, wherein in plan view of the substrate from the front, the signal processor is rectangular in shape, and the plurality of voltage terminals and the plurality of signal terminals are all provided on one side of the signal processor.
5. The optical concentration measuring device according to claim 1, wherein a portion of the wiring

connecting the first voltage terminal and the second voltage terminal is common to a portion of the wiring connecting the second voltage terminal and the third voltage terminal.

6. The optical concentration measuring device according to claim 1, wherein a wiring length from the signal processor to the first detector is shorter than a wiring length from the signal processor to the second detector.

7. The optical concentration measuring device according to claim 1, wherein a portion of the wiring has a disconnected section, and a length of the disconnected section is $1/10$ or less of a length of a signal line connecting to the detector from the signal terminal.

8. A signal processor useable in the optical concentration measuring device according to claim 1.
